Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/23/2013
PI Name:	Dhir, Vijay Ph.D.		
Project Title:	Flow Boiling Of Well Wetting Liquids Under Reduced Gravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90095-8357	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	05/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2013 per NSSC information (Ed., 6/1/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Warrier, Gopinath (University of California, Los Angeles)		
Grant/Contract No.:	NNX09AI68G		
Performance Goal No.:			
Performance Goal Text:			
	The dynamics of a single bubble under different levels of bulk liquid velocity, surface orientation, contact angle, liquid subcooling, and surface materials are studied in this work. A microfabricated cavity at the center of a flat heating surface is used to generate a single bubble. Silicon and aluminum are used as surface materials. The investigated bulk liquid velocities range from 0.076 m/s to 0.25 m/s, while surface orientation varies from horizontal to vertical, through 30o, 45o and 60o, in upflow conditions. Bulk liquid temperature is set close to saturation temperature, with bulk liquid subcooling less than 1 oC, and wall superheat is maintained between 5.2 oC and 5.9 oC. In all cases, under flow conditions, the bubble tilts in the direction of flow, departs from the nucleation site, slides along the heater surface before lifting off from the surface. It is observed that departure and lift off are governed by bulk liquid velocity, orientation of heater surface and contact angle. Bubble diameter and bubble sliding velocity at lift off are strongly affected by bulk liquid velocity and surface orientation. The dependence of lift off on static contact angle is important,		

Task Description:	with lift off diameter and time decreasing with decrease in contact angle. It is found that irrespective of orientation and contact angle, departure and lift off diameters decrease with increase in bulk liquid velocity, and that at a given velocity, orientation has almost no effect on departure diameter. Also, contact angle appears to have no effect on departure diameter. On the other hand, decrease in contact angle leads to decrease in lift off diameter. It is shown that for given conditions, bubble lift off diameter increases when the gravity component normal to the heater is decreased. Additionally, the effect of gravity on lift off diameter decreases as the liquid bulk velocity increases. Experimental results show that bubble sliding distance increases with increase in bulk liquid velocity and decrease in the magnitude of the component of gravity normal to the heater. For given conditions, sliding distance decreases with decrease in contact angle. Regarding bubble lift off diameter, it is found that it increases with increase in flow velocity, as well as with increase in component of gravity parallel to heater surface. With increase in component of gravity parallel to the surface, bubble velocity becomes larger than bulk liquid velocity. For velocities approaching 0.30 m/s, the bubble lift off diameters become independent of orientation of the surface or level of gravity. Thus the interacting effects of both gravity level and inertia are only present for velocities less than 0.30 m/s.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a study designed to enhance our fundamental knowledge of bubble dynamics and the associated heat transfer in reduced gravity environments. The knowledge gained is applicable to the design of better two-phase cooling systems such as electronic cooling, powerplants, etc. at Earth normal gravity. The mechanistic model that will be developed will serve as a tool for designing better two-phase cooling system under varying gravity levels.
Task Progress:	Several key observations can be made based on the research conducted: • Irrespective of orientation and contact angle, departure and lift off diameters decrease with increase in bulk liquid velocity. • At a given velocity, orientation has almost no effect on departure diameter. • Contact angle has no effect on departure diameter. On the other hand, decrease in contact angle leads to decrease in lift off diameter. • For given conditions, bubble lift off diameter increases when the gravity component normal to the heater is decreased. The effect gravity on lift off diameter is reduced at high liquid bulk velocities. • The bubble sliding distance increases with increase in bulk liquid velocity and decrease in the magnitude of the component of gravity normal to the heater. • For given conditions, sliding distance decreases with decrease in contact angle. • The bubble velocity at lift off increases with increase in flow velocity, as well as with increase in component of gravity parallel to heater surface. With increase in component of gravity parallel to the surface, bubble velocity becomes larger than bulk liquid velocity. • Increase in component of gravity parallel to the surface and decrease in contact angle lead to higher bubble velocity at lift off, for similar conditions.
Bibliography Type:	Description: (Last Updated: 08/23/2012)